

Exercise 7: CLASSIFICATION –BAYESIAN NETWORK

1. Build Bayesian network model using existing default data
2. Visualize Tree Augmented Naïve Bayes model
 - Naïve Bayes Classifier is one of the simple and most effective Classification algorithms which helps in building the fast machine learning models that can make quick predictions.
 - It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Where,

P(A|B) is Posterior probability: Probability of hypothesis A on the observed event B.

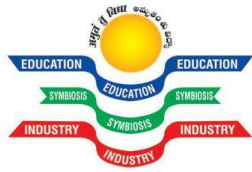
P(B|A) is Likelihood probability: Probability of the evidence given that the probability of a hypothesis is true.

P(A) is Prior Probability: Probability of hypothesis before observing the evidence.

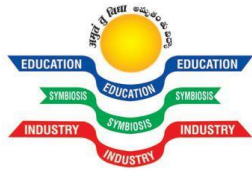
P(B) is Marginal Probability: Probability of Evidence.

PROGRAM:-

```
from sklearn import metrics
from sklearn.naive_bayes import GaussianNB
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
dataset=pd.read_csv("Iris.csv")
X=dataset.iloc[:,4].values
Y=dataset['Species'].values
```



```
X_train,X_test,Y_train,Y_test=train_test_split(X,Y,test_size=0.3)
classifier=GaussianNB()
classifier.fit(X_train,Y_train)
print(X_test[0])
y_pred=classifier.predict(X_test)
print(y_pred)
accuracy=accuracy_score(Y_test,y_pred)
print("Accuracy:",accuracy)
```



Exercise 8: CLUSTERING – K-MEANS

Write a python program

1. To perform Preprocessing
2. To perform clustering using k-means algorithm to cluster the records

Program:

```
import matplotlib.pyplot as plt

import numpy as np

from sklearn.cluster import KMeans

X=np.array([[1,1],[1.5,2],[3,4],[5,7],[3.5,5],[4.5,5],[3.5,4.5]])

print(X)

plt.scatter(X[:,0],X[:,1])

kmeans=KMeans(n_clusters=2)

kmeans.fit(X)
```